

DAY 5

1. List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

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1. Motherboard

The main circuit board that connects all components like CPU, RAM, storage, and GPU, allowing them to communicate.

2. CPU (Processor)

The brain of the computer that performs all calculations and executes instructions.

3. RAM (Memory)

Temporary memory that stores active data and programs for fast access while the system is running.

4. SMPS (Power Supply Unit)

Converts AC power from the wall into regulated DC power for all internal components.

5. Storage Drive (HDD/SSD/NVMe)

Stores the operating system, applications, and user data permanently.

6. GPU (Graphics Card)

Handles graphics processing for games, videos, and visual applications (may be integrated or dedicated).

7. CPU Cooler (Heatsink/Fan or Liquid Cooling)

Removes heat from the CPU to prevent overheating and maintain performance.

8. Cabinet/Case

The outer structure that holds and protects all internal components.

9. Cooling Fans

Help maintain airflow inside the system to keep components cool.

10. Expansion Cards (Optional)

Additional cards like sound cards or Wi-Fi cards that add extra features to the system.

2. Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step. Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.

1. Power Down & Unplug

- Action: Turn off the PC and unplug the main power cable from the wall outlet and the Power Supply Unit (PSU). Disconnect all external peripherals (monitor, keyboard, mouse).

2. Remove the Side Panel

- Action: Lay the desktop flat on a clean, non-carpeted surface. Unscrew the thumbscrews or standard chassis screws securing the left side panel, slide the panel toward the back, and lift it off.

3. Disconnect Internal Cables

- Action: Unplug all power and data cables attached to the motherboard, drives, and graphics card. Press the locking tabs on connectors (like the 24-pin main power, SATA cables, and PCIe cables) before gently pulling them straight out.

4. Remove RAM (Memory) Modules

- Action: Push down on the plastic retaining clips at both ends of the DIMM slots. The RAM module will pop up slightly; pull it straight up out of the slot.

5. Remove HDD / SSD (Storage)

- Action: Unscrew the storage drive from the drive bay or unlatch the toolless locking mechanism, and slide the drive out of the chassis.

6. Remove the Motherboard

- Action: Unscrew the screws securing the motherboard to the metal standoffs inside the case. Carefully lift the motherboard straight up and out of the chassis.

3. Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

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Motherboard Preparation: Install the CPU, RAM, and M.2 SSD onto the motherboard *outside* the case. This prevents clearance issues and damage from tight spaces. [[1](#), [2](#), [3](#)]

Power Supply (PSU) & Case: Mount the PSU in the case and route the necessary cables. It's easiest to do this before the board blocks cable routing channels. [[1](#), [2](#), [3](#)]

Motherboard Installation: Secure the prepared motherboard into the case. [[1](#), [2](#)]

Cooler Installation: Attach the CPU cooler (and connect its fan to the CPU_FAN header). *Note: Do this before installing large GPUs so you have room to screw it in.* [[1](#), [2](#)]

GPU Installation: Seat the graphics card into the top PCIe slot and lock the retention tab. Do this last as it physically blocks access to lower motherboard headers. [[1](#), [2](#)]

Front Panel & Power Cables: Connect the case power/reset buttons, front USBs, and plug in the 24-pin ATX, CPU, and PCIe power cables.

Post-Build Test: Plug in the monitor to the GPU (not the motherboard) and turn it on to confirm POST before final cable management.

4. Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

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1. Power OFF and Unplug System

Why important: Prevents electric shock and protects components from short circuits.

2. Use Anti-Static Precautions (Wrist Strap / Grounding)

Why important: Stops static electricity from damaging sensitive parts like CPU, RAM, and motherboard.

3. Work on a Clean, Dry Surface

Why important: Avoids dust, moisture, and accidental damage to components.

4. Avoid Working on Carpeted Areas

Why important: Carpets increase static electricity buildup, which can harm PC parts.

5. Handle Components Carefully (Do Not Touch Pins/Contacts)

Why important: Finger oils and pressure can damage delicate connectors and circuits.

6. Keep Screws and Small Parts Organized

Why important: Prevents loss of screws and ensures correct reassembly.

7. Ensure Proper Lighting

Why important: Helps clearly see small connectors and avoids incorrect installation.